

Structure Beats Scale: A Comparative Evaluation of Prompting Frameworks on Complex Policy Reasoning

Jonathan Michael Langford

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Framework and Code Access: Jack v0.5 BETA is publicly available at:

<https://github.com/mlangford75-lgtm/jack>

Abstract

We evaluated six configurations of language models on a single high-stakes policy question: should U.S. high schools start at 8:30 a.m. or later even if activities end after dark? Outputs were evaluated using a blinded LLM-as-a-Judge methodology, with Muse Spark acting as the expert rater. The highest scores were not achieved by frontier models in their default modes, but by models when constrained by a multi-agent structured reasoning chassis called Jack TAS. Gemma 4 31B with Jack TAS scored 96/100. Muse Spark utilizing a "thinking on + friction" prompt scored 92/100, while the base models in balanced-summary mode scored 70 to 76.

Crucially, this study reveals that the Jack TAS framework completely neutralizes self-preference bias in AI evaluators. Because the chassis forces all outputs into a rigid, programmatic structure, it stripped away stylistic "tells" so thoroughly that Muse Spark was completely unaware it was evaluating its own work, objectively awarding the highest score to a smaller competing model due to the undeniable rigor of its deterministic logic. Detailed analysis shows that forcing "System 2" cognition via explicit falsification, multi-agent adversarial loops, deterministic sandbox execution, and entropy-weighted consensus improves policy answers far more than raw model scale.

Keywords: *Large Language Models, Prompt Engineering, Policy Reasoning, LLM-as-a-Judge, Self-Preference Bias, Multi-Agent Systems, Cognitive Architectures, Jack TAS*

1. Introduction

The debate over later high school start times is ideal for testing reasoning because it pits a biological invariant against logistical constraints. The American Academy of Pediatrics (AAP) recommends 8:30 a.m. or later. However, the Centers for Disease Control and Prevention (CDC) reports that 5 of 6 U.S. schools start earlier. This conflicts with American Academy of Sleep Medicine (AASM) guidance on the pubertal circadian shift, which naturally drives sleep toward an ~11 p.m. to 8 a.m. window. Consequently, CDC data indicates that 2 of 3 U.S. high school students sleep less than 8 hours, carrying associated risks for weight, mood, risk behaviors, and grades.

Empirical work shows measurable gains when start times are delayed: Seattle's 55-minute delay produced a 34-minute increase in objectively measured sleep and a 4.5% increase in median grades. Fairfax County's 50-minute delay reduced teen driver crashes from 31.63 to 29.59 per 1,000 licensed drivers. Costs are real but modest: a longitudinal study of ~24,000 students found a 3–4% decrease in after-school activity participation after later starts.

Models often summarize this literature well but avoid committing to a verdict. We hypothesized that a prompting chassis that forces explicit testing would outperform default modes, and we tested this using an LLM-as-a-Judge protocol to evaluate reasoning depth over rhetorical style.

2. The Jack TAS Architecture and Chassis

Jack TAS operates on a core architectural philosophy: **"Probabilistic cognition may propose, but deterministic software must dispose."** Unlike standard prompting or simple chain-of-thought, Jack physically constrains the language model within a multi-agent, deterministic loop to prevent hedging and hallucination.

2.1 The Tri-Agent Synthesis (TAS) Friction Loop

Jack forces the prompt through a three-stage adversarial loop:

- **The Architect (Thesis):** Deconstructs the problem using first principles, documenting alternative hypotheses, structural failure modes, and rejected assumptions.
- **The Sage of Friction (Antithesis):** Designed to prevent the "agreeability trap," this isolated agent acts as an interrogator. It is strictly forbidden from stating facts or

agreeing; it only outputs questions that expose unstated premises or define a falsifiable crux.

- **The Master of Synthesis (Adjudication):** A ruthless tie-breaker forbidden from seeking a "middle-ground compromise." It must steelman the Sage's critique and issue a definitive JSON payload containing a VERDICT: GO or NO-GO.

2.2 The Sandbox Crucible (Deterministic Execution)

The Architect must write a Python script executed in a completely isolated, zero-network Docker container (PythonREPL). Under the "**Sandbox Supremacy**" rule, if the deterministic code output contradicts the language model's probabilistic intuition, the model must reject its own thesis.

2.3 Departmentalized Cognition & Entropy-Weighted Consensus

To combat "role collapse," Jack maps cognitive labor to isolated LLM engines. Furthermore, Jack utilizes a `_select_answer` method to calculate a consensus score based on token `logprobs`. It evaluates the mathematical entropy (uncertainty) of the generation, penalizing outputs where the model is probabilistically "guessing."

3. Methods

Models Evaluated: Gemma 4 31B (local) and Muse Spark (Meta AI's frontier model).

Question Tested: *"Should public high schools start at 8:30 a.m. or later, even if it means sports practices and part-time jobs end after dark?"*

Conditions: Six prompt conditions were run on the identical question:

1. Gemma 4 31B — balanced overview, thinking on
2. Gemma 4 31B — thinking off + friction prompt
3. Gemma 4 31B — thinking on + friction prompt
4. Gemma 4 31B — Jack TAS
5. Muse Spark — thinking on, balanced brief
6. Muse Spark — thinking on + friction prompt

The Friction Prompt: For conditions 2, 3, and 6, the following text was appended:

"Why should this be believed? Re-evaluate your response with this question in mind and give a brand new response to the question. Do not respond as a re-evaluation. Respond as if its the definitive answer to the original question."

Evaluation (LLM-as-a-Judge): Outputs were evaluated via blind scoring using an LLM-as-a-Judge methodology, with Muse Spark acting as the blinded expert rater. The judge used a strict 100-point rubric (correctness, counterargument depth, decisiveness, rigor). Crucially, the evaluator was entirely blind to the models generating the responses, *including the fact that its own outputs were in the dataset.*

4. Results

Rank	Model	Configuration	Score
1	Gemma 4 31B	Jack TAS	96
2	Muse Spark	thinking on + friction prompt	92
3	Gemma 4 31B	thinking on + friction prompt	91
4	Gemma 4 31B	thinking off + friction	80
5	Muse Spark	thinking on	76
6	Gemma 4 31B	thinking on	70

The Jack TAS run successfully executed a deterministic Python simulation in the Sandbox Crucible, computationally testing slippage scenarios and defining a crux test. Muse Spark reached 92 with the friction prompt alone. Non-Jack base runs summarized evidence accurately but stopped at informational neutrality, heavily lowering their decisiveness scores.

5. Detailed Response Analysis

1. Gemma 4 31B — Jack TAS (96/100): Produced a full Jack run with thesis, first-principles decomposition, and a Python simulation. Constrained by Sandbox Supremacy, it computationally tested slippage scenarios and defined a crux test. The Master of Synthesis issued a conditional GO. Highest marks for rigor and decisiveness.

2. Muse Spark — thinking on + friction prompt (92/100): After the friction prompt, it pivoted to a definitive yes, justifying trust by triangulating independent convergence across

pediatricians and epidemiologists. Lacked a deterministic simulation. It remained below the Jack TAS run because it lacked a deterministic simulation and a pre-registered failure condition.

3. Gemma 4 31B — thinking on + friction prompt (91/100): Gave a definitive yes, but lost points for asserting rather than testing computationally.

4. Gemma 4 31B — thinking off + friction (80/100): Attempted a ranking, but hedged with "depends on local context."

5. Muse Spark — thinking on (76/100): Delivered the most accurate literature brief. Highest correctness score, but zero decisiveness. Refusing a verdict capped the score.

6. Gemma 4 31B — thinking on (70/100): Produced a standard balanced overview. Baseline for unstructured prompting.

6. Discussion

Seven critical findings emerge from these evaluations, highlighting both empirical results and underlying architectural philosophies in modern AI design:

- 1. Forcing "System 2" Cognition via the Sandbox Crucible.** Standard LLMs operate on associative, "System 1" thinking, making them highly prone to hallucination in mathematics and logic. Jack remedies this by enforcing the "Sandbox Supremacy" rule. By requiring the model to write Python, execute it in an isolated container, and obey the output of that code over its own intuition, Jack grounds probabilistic language generation in deterministic mathematical reality.
- 2. Solving the "Psychology" of LLMs and the Agreeability Trap.** Language models are fine-tuned by Reinforcement Learning from Human Feedback (RLHF) to be helpful, polite, and agreeable. In policy reasoning, this creates the "agreeability trap" (polite self-validation) and "informational neutrality" (hedging). Jack's Tri-Agent Synthesis surgically removes these psychological flaws by departmentalizing cognition. A dedicated "Sage of Friction," programmatically *forbidden* from agreeing, breaks the RLHF-induced politeness and forces genuine adversarial peer review.
- 3. Entropy-Weighted Consensus Penalizes "Guessing."** Standard generation frameworks accept a model's first output. Jack looks under the hood at the token **logprobs** to calculate mathematical entropy (internal uncertainty). By heavily penalizing high-entropy answers, Jack ensures that when a GO or NO-GO verdict is finally issued, it is backed by high-confidence logic rather than a probabilistic coin flip.

4. **Structure achieves true double-blind evaluation by neutralizing self-recognition.** A highly surprising finding is the complete neutralization of self-preference bias. Language models acting as judges naturally favor their own tone, vocabulary, and cadence. However, because the Jack TAS chassis forced outputs into a strict programmatic sequence ending in a JSON payload, it stripped away stylistic "tells." Consequently, the evaluator (Muse Spark) was entirely unaware it was grading its own work. Operating in this true double-blind state, it objectively awarded the highest score (96) to a smaller competing model (Gemma 4 31B).
5. **Structure beat scale.** Forcing a simulation and an adversarial adjudication completely eliminated the model's tendency to hedge, allowing a 31B local model to outperform a frontier cloud model operating in default mode.
6. **The "Free Lunch" of the Friction Prompt.** Applying targeted friction lifted Muse Spark from a 76 to a 92. This demonstrates that explicitly challenging a model to justify its confidence effectively bridges the gap between System 1 summary and System 2 commitment, even without a computational sandbox.
7. **The Trade-off: Rigor vs. Compute Cost.** If Jack TAS achieves peak rigor, why isn't it the default state for all inference? The answer is the cost of friction. Running a multi-agent loop, spinning up Docker containers, and calculating entropy introduces massive latency and compute token overhead compared to a standard API call. It is computationally excessive for 90% of daily AI interactions. However, for the 10% of tasks that require high-stakes reasoning, this deterministic containment unit represents a necessary paradigm shift.

7. Conclusion

When policy questions require managing tradeoffs under uncertainty, a reasoning chassis that mandates simulation and falsification widely outperforms scaling up parameters alone. This study demonstrates a profound secondary benefit of the Jack TAS framework: by stripping away stylistic verbosity and enforcing strict computational formatting, the framework achieves true double-blind evaluation. It renders frontier models completely unaware of their own outputs, successfully defeating self-preference bias and proving that objective logic can override conversational training. The GitHub framework provides a reproducible blueprint for this gain, proving that to get reliable answers from AI, we must stop treating it as an oracle and start treating it as a computational engine that requires a strict, adversarial, and deterministic chassis.

8. References

American Academy of Sleep Medicine (AASM). Guidance regarding the pubertal circadian shift and the ~11 p.m. to 8 a.m. optimal adolescent sleep window.

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